

What Substitutes for a Stale Mark? Breadth, Implied Volatility, and the Price of Being Late

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Abstract

Anyone forecasting the risk of an asset that is priced infrequently faces the same dilemma. The asset's own history is the most informative thing about it, but that history has gone stale, while liquid markets have kept trading. What can be bought instead of timeliness? We run the experiment where the truth is observable, handicapping a forecaster of S&P 500 realised variance by δ days while other sources stay current, and we price two candidate substitutes against each other. Foreign index closes recover a rising share of the loss, from nothing at one day to 72% at eleven weeks — 72% measured by R^2 on log normalised variance, which is the paper's headline scale, and 81% under QLIKE on the variance scale, which is the proxy-robust one. Both figures are stated throughout because the scale changes the number. Those rates are the prespecified seven-regressor estimate; Section 5.1 shows that compressing the foreign block into a single real-time factor forecasts better and raises $R(55)$ to 75.2%, inside the interval the seven-regressor rate already carries, so the headline is not restated at the larger figure. Implied volatility recovers more than all of it: with VIX and VDAX in hand, a forecaster three weeks stale beats one holding fully current domestic data. Given implied volatility, the foreign cross-section then adds nothing at any delay, and once we measure what carrying seven extra regressors costs, its incremental content is indistinguishable from zero rather than negative. Both results hold whether VIX is lagged to respect closing times or used same-day in the objection's favour. The practical reading is narrow and, we think, useful: where a liquid options market exists on the stale asset, breadth is redundant, so the cross-sectional result may be most relevant to assets without such a market — the case that motivates the question and the one this experiment cannot itself demonstrate.

1. The question

A private equity portfolio is marked once a quarter. Direct property is appraised on a cycle, and the resulting series is smoothed in ways that have been documented for thirty years [8]. Hedge fund returns arrive with a serial correlation that understates their true volatility [7]. In each case someone must produce a risk number today from marks that are weeks or months old, while public markets have gone on trading in plain sight.

The instinct is to reach for those markets. The question is which of them, and how far it gets you. Two candidates present themselves. The first is breadth: other markets are correlated with yours, and some of them have already closed today. The second is the options market, which reports a forward-looking risk number continuously and without any delay at all. These are different purchases, and a practitioner choosing between a wider data feed and an options subscription would like to know which one buys back more of what staleness took.

We cannot observe the truth for a private portfolio, so we run the experiment where the truth is observable. We take a forecaster of S&P 500 realised variance, withhold their own data for δ days, leave the other sources untouched, and measure what each gives back. For foreign equities the asymmetry costs nothing to construct: Tokyo closes at 06:00 UTC, Hong Kong at 08:00, Mumbai at

10:00, London and Frankfurt at 16:30, New York at 21:00. On any given day the rest of the world has already finished trading when New York does.

2. What is already established

Four findings underpin this exercise, and none of them is ours.

Foreign equity volatility forecasts domestic volatility. Engle, Ito and Lin rejected the idea that volatility is a purely local phenomenon in 1990 [3], and the spillover literature since has measured the connections in considerable detail [2, 4]. Korkusuz augments a HAR model of S&P 500 volatility with a European block and improves on it [9]. Buncic and Gisler run the mirror experiment, forecasting seventeen foreign markets from US volatility, and report out-of-sample R^2 between 4.6% and 14.5% [1].

Non-synchronous closing times define which observations a forecaster may legitimately use. Zhao, Qiao and Lin say so explicitly and build an attention model whose mask prevents a closed exchange from acting as a source [15]. Jayawardena and co-authors forecast Tokyo volatility from markets that trade while Tokyo sleeps [6].

A live market can price a stale one. Goetzmann, Ivković and Rouwenhorst showed that same-day US returns predict the next day's international fund NAVs strongly enough to trade on [5], which is the mechanism of this paper stripped to its simplest form.

Implied volatility carries much of the cross-market gain. This is the finding that shapes the design below. Both Korkusuz [9] and Buncic and Gisler [1] report that a large share of what looks like a cross-market effect is attributable to implied volatility, the VIX in particular. Their work does not settle the matter for a *delayed* forecaster, since in both cases the domestic information set is current, but it makes the comparison unavoidable: any claim that foreign markets substitute for timeliness has to be set against an options market that never goes stale.

What none of this work does is treat the delay as a variable. In every case additional data is added to an information set that is already current, and the question is whether it helps. Ours is the complementary question: when domestic information is not current, how much of the damage does each candidate undo, and how does that fraction move as the damage grows? That is a rate of substitution between things a practitioner can actually buy, and we could not find it measured.

3. How we measure it

The data are daily open, high, low and close for eight equity indices — S&P 500 (SPX), Nikkei 225 (N225), S&P/ASX 200 (AXJO), Hang Seng (HSI), Nifty 50 (NSEI), FTSE 100 (FTSE), DAX (DAX) and Bovespa (BVSP) — from January 2000 to September 2026, together with two implied-volatility indices, CBOE VIX and VDAX-NEW. All series are retail end-of-day exports from Investing.com, downloaded in September 2026; the repository lists every file, the identifier it maps to, and the column dictionary, and pins the environment to NumPy and pandas alone. Each series came in two exports, because the source caps a download at 5,000 rows, which leaves ten joins to be checked rather than trusted. Nine of the ten overlap by 120 to 251 trading days, and across every overlapping row the two exports agree exactly: no mismatches, largest absolute difference zero. The tenth is the S&P itself, where the exports abut without overlapping and no cross-check is possible. We report it as unverified and fall back to two weaker tests: the join is contiguous, Friday 9 March 2018 to Monday 12

March 2018 with no session missing or duplicated, and the return across it is -0.0013 , or 0.11 standard deviations of the series' own daily returns, so there is no level shift where the exports meet.

A foreign close is admissible only when its timestamp falls before the target's. Markets closing at or after the target therefore contribute their previous session. The rule generalises without special cases, which matters for Section 8: a Tokyo forecaster gets every other market a day late, because Tokyo closes first. It also disciplines the implied-volatility block, and the discipline is awkward in a way worth setting out precisely. VDAX-NEW satisfies the rule outright, since Frankfurt finishes hours before New York. VIX does not. Its printed daily close is stamped 21:15 UTC, fifteen minutes after the 21:00 S&P cash close, so that print is inadmissible. VIX is computed continuously from live SPX option quotes, so a value of it did exist at 21:00 and would be the right observation to use, but a retail end-of-day export carries no intraday snapshot and we do not have it. What we can run is a pair of timing specifications on either side of the one we want: the previous day's close, which is strictly admissible and discards a full day of information, and the same-day close, which contains fifteen minutes we are not entitled to. Section 7 leads with the lagged arm, because it is the one that obeys our own rule, and reports the same-day arm as a sensitivity check. The 21:00 observation is temporally between the two, and we are careful not to infer from that that its forecasting performance would be numerically between their results. Timestamps order; skill need not.

Volatility is the Yang-Zhang estimator [14] over rolling five-day windows, which uses the whole daily range rather than closing prices alone. For each index we work with $a_t = \log(\sigma^2_{YZ,t} / M_t)$, where M_t is the trailing 252-day median, and the implied-volatility series enter the same way, as the log of the level over its own trailing median. Dividing by a series' own median puts Tokyo, São Paulo and the VIX on one scale and keeps each series' units out of the regression. One detail of that definition needs stating, because it looks like look-ahead and is not. The target a_{t+5} carries M_{t+5} , a median whose window includes the five days between the forecast origin and the outcome, which a forecaster at t does not know. No feature is affected — every one is dated $t - \delta$ or earlier for the domestic block and t for the others — but the quantity being predicted is not one a practitioner could convert back to a variance forecast at the moment they make it. The implementable alternative normalises the outcome at the ORIGIN, $\log(\sigma^2_{YZ,t+5} / M_t)$, and we ran the whole experiment both ways. The two targets differ by the five-day drift in a 252-day median, which has a standard deviation 3.7% of the target's own and correlates +0.52 with it, so it is neither large nor independent. It changes nothing: the substitution rate is 71% against 72% at eleven weeks, and no delay differs by more than a point. We keep the target-dated definition and report the origin-dated run as the check that licenses it. The target is a_{t+5} , five trading days ahead, left continuous: a companion note [13] showed that splitting such a target at its median throws away exactly the magnitude information a cross-sectional model would use, so we do not split it.

Models are built from three feature blocks. The domestic block is the target's own history at daily, weekly and monthly aggregation, in the manner of Corsi [2], observed at $t - \delta$. The foreign block is seven index decision variables at their admissible timestamps — the six that close before New York, plus Bovespa a session late. The implied-volatility block is VIX and VDAX. Only the domestic block is delayed; the others are what a forecaster could see today, which is the whole point of the comparison. Every model is a ridge regression refitted every 21 days on a rolling 1,250-day window, with the penalty chosen on a 250-day validation tail and features standardised on training statistics alone. All are trained on the same labels — a pair is admitted only once its outcome was resolvable at $t - \delta$ — so

differences between them are features and nothing else. The rolling fixed window is not incidental; it is what makes the tests in Sections 6 and 7 valid.

Losses need care here, and an earlier draft of this paper got the justification wrong. With an imperfect volatility proxy only a restricted family of losses preserves the ranking a researcher would obtain against true latent volatility [17], and that family is defined on the VARIANCE scale. Our target is a log variance, so the two halves of the usual pairing land differently. QLIKE survives the transformation: it depends on the ratio of outcome to forecast alone, and dividing both by the same trailing median leaves that ratio untouched, so what we compute is exactly the variance-scale QLIKE. Squared error does not. Computed on log variance it is outside the family, because a monotone transform of the arguments is precisely what the robustness result fails to survive.

We therefore report three. Out-of-sample R^2 on the log scale, which is the paper's headline and is *not* proxy-robust. QLIKE, which is. And squared error on the variance scale, which is Patton's MSE and is also robust; reconstructing a variance forecast from a log one needs a correction, since $\exp()$ of a log forecast is median-unbiased rather than mean-unbiased and squared error punishes exactly that, so we apply Duan's smearing estimator computed from training residuals at each refit. Mean absolute error, used in some recent work in this area, is in no version of the family. Where conclusions differ across losses we give weight to what both proxy-robust losses support; where those two disagree with each other, we treat the finding as unresolved, and we say so at the one place in the paper where that happens.

Write $S(\delta)$ for skill at delay δ . Delay costs $S(0) - S_{\text{own}}(\delta)$. A candidate substitute returns $S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)$. The substitution rate is their ratio,

$$R(\delta) = [S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)] \div [S_{\text{own}}(0) - S_{\text{own}}(\delta)].$$

It has no units, so it can be compared across loss functions, across markets, and across candidates. One caution, which Section 7 will need: the measure assumes the substitute is *restoring* something delay destroyed. A source that also carries information the domestic history never had can exceed 100%, and when it does, the ratio has stopped answering the question it was built for.

4. What breadth recovers

The test window runs from May 2007 to September 2026, 4,862 days.

Table 1. Out-of-sample skill against the training mean, and the substitution rate under each loss. Skill is the fraction of benchmark loss removed, so the two pairs are comparable. Intervals bootstrap the whole ratio, recomputing numerator and denominator on the same resampled days. R^2 is on the log scale and is not proxy-robust; QLIKE is.

δ	R^2 own	R^2 cross	$R(\delta)$ on R^2	QLIKE skill own	QLIKE skill cross	$R(\delta)$ on QLIKE
0	0.4987	0.4884	—	0.5721	0.5578	—
1	0.4518	0.4512	−1% [−32, 26]	0.5196	0.5173	−4% [−43, 31]
3	0.3848	0.4262	36% [17, 54]	0.4384	0.4920	40% [16, 62]
5	0.3371	0.4202	51% [36, 66]	0.3738	0.4855	56% [37, 73]
8	0.2822	0.4111	60% [47, 71]	0.2873	0.4749	66% [50, 78]
13	0.2011	0.3907	64% [53, 73]	0.1309	0.4526	73% [59, 83]
21	0.1121	0.3726	67% [57, 76]	−0.0182	0.4358	77% [64, 86]
34	0.0404	0.3603	70% [61, 77]	−0.1366	0.4282	80% [69, 87]
55	−0.0351	0.3470	72% [64, 79]	−0.1769	0.4223	80% [69, 88]

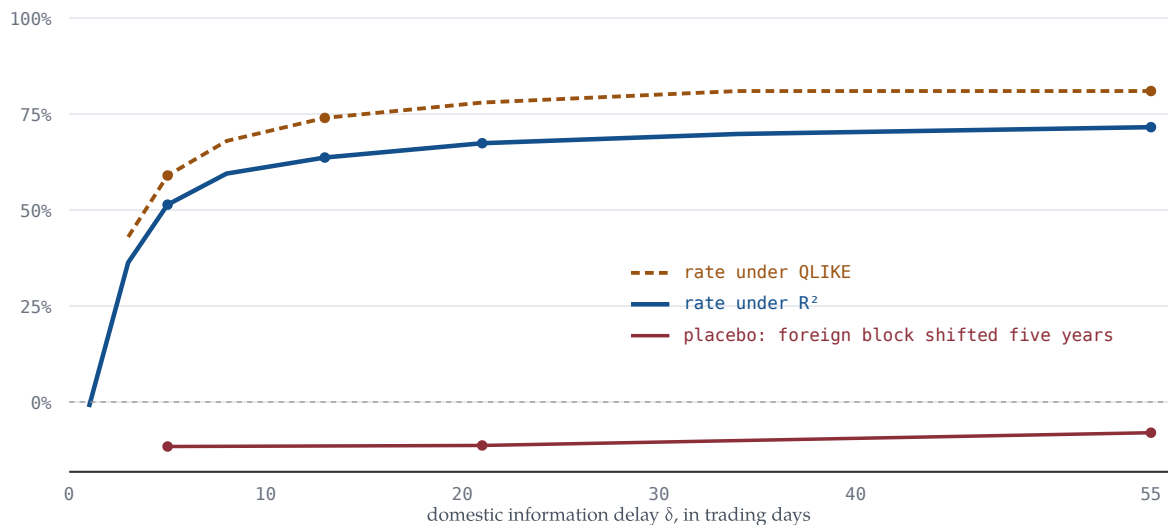


Figure 1. Breadth buys back almost nothing at one day, half the loss within a week, and around three-quarters by eleven weeks. The placebo, described in Section 6, keeps the same foreign series but destroys their alignment with the target, and the effect disappears.

Three things stand out. At one day the rate is indistinguishable from zero, which is what should happen: the S&P's own recent history is still the best available description of the S&P, and the foreign block we selected produces no measurable net improvement on it. By eleven weeks the picture has inverted. The domestic model's out-of-sample R^2 has gone negative, meaning it would be better to forecast the training mean, while the cross-sectional model still holds 0.347, about 70% of what was available with fully current data. And between those extremes the rate climbs steeply and then slows, which raises the obvious question of where it stops.

Section 3 promised a third loss, and it belongs here because it is the robust one that most resembles the headline. Under squared error on the variance scale the substitution rate reads 85% at five days and 90% at eleven weeks, against 51% and 72% on the log scale. Both robust losses therefore put the rate ABOVE the headline, which makes the number we lead with the conservative one. The variance-

scale intervals are very wide at short delays, $[-74\%, 126\%]$ at five days, because the denominator there is small and squared error on raw variance is dominated by a handful of crisis days; that is why QLIKE, not MSE, is the robust loss we lean on.

5. Why breadth stops where it does

The rate flattens near three-quarters rather than approaching one, and the reason is not subtle once measured. At very large delay the domestic model retains little usable out-of-sample skill, so the cross-sectional model is working almost entirely from foreign information. Its ceiling is therefore whatever share of the S&P's volatility the rest of the world can see.

We measured that share directly. Regressing the S&P's decision variable on the six foreign closes of the same day gives $R^2 = 0.695$. Roughly 30% of the S&P's current volatility state is not linearly spanned by those six series in this sample. We resist calling that share genuinely local, because the regression does not show it: a non-linear function of the same closes, or a wider cross-section, could reach some of it. What the number does bound is how much a LINEAR model of THESE markets can see, which is the ceiling the fitted model is working against. Predicting five days ahead from foreign closes alone gives $R^2 = 0.399$ against 0.498 from domestic history, or 80% of it, which brackets the 72% the fitted model achieves. The gap between the in-sample bound and the fitted figure is consistent with the price of estimating parameters rather than knowing them, and Section 5 measures a cost of the right order; the design does not establish that estimation is the whole of it.

The cross-section is also nearly one-dimensional. The first principal component of the six foreign volatility series accounts for 66.5% of their variation and the first two for 76.9%. Six markets are close to one global volatility factor plus idiosyncratic noise, which carries a practical implication: an eighth or a ninth correlated index may add relatively little under this linear specification, because what a linear model of these markets can see is set by how much of the S&P is global rather than by how many global series are counted.

That leaves the other half of the sentence above: what it costs to estimate parameters rather than know them. We had treated this as a residual, and it is large enough to deserve a number. Breadth is not free. Seven foreign coefficients must be estimated from a finite window, and the estimation is paid for whether or not the coefficients turn out to matter.

Two measurements agree on the size of the bill. The first varies the number of peers at $\delta = 0$ and reads off the slope: the penalty is a straight line through the origin at -0.0014 of R^2 per market, which accounts for 98% of the variation across peer counts and implies -0.0100 for the full seven. The second replaces the foreign block with AR(1) surrogates fitted to each peer, drawing innovations from a multivariate normal with the peers' own contemporaneous covariance so the surrogate cross-section is as correlated as the real one, while carrying no information about the target at any lag. Averaged over one hundred draws this gives -0.0131 at $\delta = 0$, with a Monte Carlo standard error of 0.0003, an order of magnitude below the sampling interval that follows and therefore not the uncertainty that binds. The two methods differ by 0.003, which is agreement on the magnitude rather than a formal match, and we do not read a third decimal place from either. About one point of R^2 is the price of carrying the cross-section. Section 5.2 measures it a third time, with a generator that fits nothing at all, and gets the same answer.

Two things follow. The first settles a reading we had left open. At $\delta = 0$ the measured difference is -0.0103 and the cost is -0.0131 , so the gross content of the foreign block is $+0.0028$. Gross is a difference of two estimated quantities, so we bootstrap it, resampling test days in moving blocks and recomputing the measured difference and the surrogate cost on the same days so that their errors stay correlated. The interval is $[-0.0060, +0.0115]$: indistinguishable from zero, and bounded above at about one point of R^2 . Foreign markets are not misleading about the current state of the S&P. Whatever they know about it that fully current domestic data does not already carry is too small for this design to detect, and the negative sign was the bill rather than the information.

The second is that the bill grows with delay, from -0.013 at $\delta = 0$ to -0.054 at $\delta = 21$. This is a property of the fitting procedure rather than of the markets: as domestic information goes stale the fit weakens, the validation step selects a looser penalty, and the seven regressors are given more rope. Separating the two makes the gross and net readings diverge in a way that matters for anyone deciding whether to build this.

Table 2. What the cross-section knows, and what a forecaster keeps after paying to estimate it. Cost from AR(1) surrogates, one hundred draws, innovations carrying the peers' own cross-market covariance. Gross is bootstrapped in the text: indistinguishable from zero at $\delta = 0$, marginal at $\delta = 1$, positive from $\delta = 2$.

δ	0	1	2	3	5	8	13	21
Net ΔR^2 (measured)	-0.0103	-0.0006	$+0.0188$	$+0.0414$	$+0.0831$	$+0.1289$	$+0.1896$	$+0.2605$
Cost of breadth	-0.0131	-0.0155	-0.0180	-0.0201	-0.0244	-0.0306	-0.0409	-0.0536
Gross ΔR^2	$+0.0028$	$+0.0149$	$+0.0368$	$+0.0616$	$+0.1075$	$+0.1595$	$+0.2305$	$+0.3141$

Gross content clears zero at two days, $[+0.0179, +0.0555]$, and the net figure crosses there too. At one day the gross interval is $[+0.0009, +0.0280]$, which excludes zero by a margin of 0.0009 — close enough to the boundary that we report it and decline to build on it. An earlier draft, using ten surrogate draws rather than a hundred, put that interval across zero and we withdrew the reading entirely; a better cost estimate moved it back. A band in which the foreign block holds something real that cannot yet pay for its own estimation is therefore consistent with these data at one day, and no more than that. At $\delta = 21$ estimation consumes 17% of what the cross-section knows, $+0.3141$ gross against $+0.2605$ net. That gap is not a law of nature. It is the cost of a particular estimator, and a cheaper one — a factor representation of the foreign block rather than seven free coefficients, which the principal-component result above already suggests would lose little — should recover part of it. We have not built that, and we make no claim about how much it would recover.

5.1 Fitting the cheaper representation, instead of inferring it

Two facts above sit next to each other without being joined. The foreign block is nearly one-dimensional — first principal component 66.5%, first two 76.9% — and carrying seven regressors costs about a point of R^2 at $\delta = 0$ and more as the delay grows, measured by replacing the block with surrogates that carry no signal. It is natural to conclude that a cheaper parameterisation would recover part of that cost, and an earlier draft did conclude it. That is an inference from a hypothetical generator, not a demonstration, and the demonstration is cheap.

So compress the block and compare. The eigenvectors are recomputed at every refit from the covariance of the training window alone, on columns standardised with training statistics alone, with each eigenvector's sign pinned so the factor does not flip arbitrarily between refits. A basis computed

once on the whole sample would leak: “the first component of the foreign block” is not something a forecaster in 2009 could have written down from 2026 data. An equal-weighted mean of the seven series is included as the compression that requires no estimation at all.

Table 3. Out-of-sample R^2 for four representations of the same seven foreign series, against the domestic-only model. PC1 and PC2 use one and two real-time global factors; FULL is the seven-coefficient model the rest of the paper uses. The last columns test PC1 against FULL, which are non-nested, by Giacomini-White.

δ	domestic	+ mean	+ PC1	+ PC2	+ all seven	PC1 – all seven	GW z
0	0.4987	0.4999	0.5001	0.4975	0.4884	+0.0117	2.91
3	0.3848	0.4374	0.4385	0.4358	0.4262	+0.0124	1.95
5	0.3371	0.4310	0.4325	0.4300	0.4202	+0.0123	1.87
13	0.2011	0.4005	0.4032	0.4012	0.3907	+0.0125	1.74
21	0.1121	0.3841	0.3873	0.3842	0.3726	+0.0147	1.87
55	−0.0351	0.3617	0.3664	0.3630	0.3470	+0.0194	2.08

One real-time factor beats all seven series at every delay, by 0.012 to 0.019 of R^2 , and the margin grows with delay exactly as the surrogate cost does. The gain reaches conventional significance at $\delta = 0$ and $\delta = 55$ and falls short of it in between, so the pattern is more convincing than any single entry. The real-time first component explains 66.6% of the block's variance averaged over refits, against 66.5% in the full-sample figure quoted above, which says the basis is stable enough to be estimated as you go.

Reporting a new preferred specification on the log scale alone would be inconsistent with Section 6, which exists to establish that the scale changes the answer. Under QLIKE, the proxy-robust loss, one factor wins at every delay, exactly as it does under R^2 . Under squared error on the variance scale a compression wins at five of six delays; the exception is $\delta = 0$, where the seven-regressor model scores lowest. That loss is dominated by a handful of crisis days, which is why Section 6 reports its intervals as uninformative at short delays, so the exception is a statement about tails rather than about information. Where the three losses appear to disagree elsewhere they are choosing between one factor and two, which differ by thousandths and are not a disagreement about anything.

The compression also raises the substitution rate, and a referee has asked which number the paper is claiming. Recomputed with the same full-ratio bootstrap as the headline, $R(55)$ is 71.6% [63.6%, 78.8%] for the seven-regressor model, 74.3% [67.2%, 80.9%] for the equal-weighted mean and 75.2% [68.2%, 81.8%] for one factor. Every compressed estimate sits inside the interval the seven-regressor model already carries, so the two are not distinguishable and we do not replace the headline with the larger figure. We report 72% as the prespecified full-block estimate, note the compressed estimate beside it, and say plainly that the difference is inside the noise of both.

Two qualifications, and the first is the more important. The equal-weighted mean captures nearly all of the gain: PC1 beats it, but by 0.001 to 0.005. What is being bought is the decision to compress, not the machinery of compressing, and a practitioner should take the average rather than the eigenvector. Second, this changes the $\delta = 0$ row. The seven-coefficient model scores 0.4884 against the domestic model's 0.4987 — the small negative this paper has declined to interpret — while PC1 scores 0.5001, above it. We do not claim the cross-section helps at $\delta = 0$ on that basis: it is a difference of 0.0014 with no interval attached, and reading an uninterval-ed difference as a finding is the specific error this

paper has made and corrected three times. What the row supports is narrower and sufficient, that the negative at $\delta = 0$ is a property of the parameterisation rather than of the information.

5.2 Is the bill an artefact of the generator?

Both measurements of the bill are relative to something. The slope method extrapolates from peer counts; the surrogate method replaces the foreign block with fitted Gaussian AR(1) series. A referee objects to the second, correctly, that it preserves first-order persistence and contemporaneous cross-market covariance and nothing else — not heavy tails, not volatility-of-volatility, not crisis clustering, not higher-order serial structure. Under that objection the cost is identified only relative to a generator we chose, and a forecaster's real regressors are not Gaussian AR(1) series.

The objection has a direct test. Measure the bill again with a generator that fits nothing at all: take the real foreign block and move it circularly through the sample. Every marginal, every tail, every cluster is then the real one, and the only thing destroyed is the alignment with the target. If an unfitted generator returns the fitted generator's number, the objection is answered rather than conceded.

Table 4. The cost of breadth under a fitted and an unfitted null, on identical days. AR(1) is the generator used above, twenty draws; SHIFT is the real foreign block moved circularly with its real missing-data mask held fixed, twenty offsets, with the Monte Carlo standard error of that mean beside it. NAIVE is the same shift applied to the raw array, so the holidays move with the values; it is reported as a diagnostic, not as a candidate.

δ	net ΔR^2	cost, AR(1)	cost, SHIFT	(s.e.)	cost, NAIVE	gross, AR(1)	gross, SHIFT
0	−0.0103	−0.0125	−0.0122	0.0009	−0.0329	+0.0021	+0.0019
1	−0.0006	−0.0148	−0.0140	0.0010	−0.0331	+0.0142	+0.0134
3	+0.0414	−0.0198	−0.0176	0.0013	−0.0342	+0.0612	+0.0590
5	+0.0831	−0.0242	−0.0209	0.0016	−0.0355	+0.1073	+0.1040
13	+0.1896	−0.0409	−0.0349	0.0027	−0.0436	+0.2306	+0.2245
21	+0.2605	−0.0546	−0.0487	0.0039	−0.0519	+0.3151	+0.3092
55	+0.3821	−0.0701	−0.0694	0.0071	−0.0683	+0.4522	+0.4515

One bookkeeping point before the result, because the table invites a fair objection. The AR(1) column here is twenty draws on this subsection's delay grid, not the hundred behind Table 2, so the two tables give slightly different figures for the same quantity: −0.0125 against −0.0131 at $\delta = 0$, and −0.0546 against −0.0536 at $\delta = 21$. That spread is Monte Carlo error and nothing else. Table 2 remains the paper's estimate of the bill, because it rests on more draws. What Table 4 is for is the comparison *within* it, between two columns computed on identical days from an identical number of draws, and that comparison is unaffected by how many draws either column uses.

The two generators agree at every delay. Compared against three times its own combined Monte Carlo standard error — delay by delay, rather than against the loosest tolerance in the table — the largest gap is 0.0060 of R^2 at $\delta = 13$, which is 0.60 of the tolerance there, and no delay comes closer. The gross readings move by less: +0.0021 becomes +0.0019 at $\delta = 0$, and +0.3151 becomes +0.3092 at $\delta = 21$. Nothing in Section 5 depends on which generator supplied the number, and that is the sense in which the referee's objection is answered. That claim is worth only as much as the null generator behind it, and an earlier version of this measurement was wrong twice before it was right. Appendix B records both mistakes and what they cost, because a placebo that quietly stops being a placebo returns a plausible number with no outward sign of trouble.

6. Does it survive testing

Bootstrap intervals describe a difference; they do not test a hypothesis, and our domestic model is nested inside our cross-sectional one, which rules out the usual comparison. Diebold has written at length about the misuse of his own test in exactly this situation [12]. We use Giacomini-White [10] as the primary test, because it is valid for nested models and, being loss-agnostic, covers the QLIKE result as well as the squared-error one. It requires a fixed rolling estimation window, which is what Section 3 describes. Clark-West [11] serves as a second opinion on squared loss, with a qualification developed in Section 6.1 that is the reason it is the second opinion and not the first.

Table 5. Test statistics, HAC-adjusted with a Bartlett kernel at nine lags. Positive favours the cross-sectional model. All entries have two-sided $p \leq 0.011$; all except QLIKE at $\delta = 3$ have $p < 0.01$.

δ	3	5	8	13	21	34	55
Giacomini-White, squared loss	3.18	4.48	5.08	5.34	5.66	6.21	7.02
Giacomini-White, QLIKE	2.57	3.18	3.24	2.77	2.87	3.00	3.28
Clark-West	7.09	8.02	8.28	8.09	8.01	8.51	9.21

P-values are two-sided throughout. On that basis every statistic from $\delta = 3$ onward has $p \leq 0.011$, and all but one are below 0.01: the exception is QLIKE at $\delta = 3$, where $z = 2.57$ gives $p = 0.0102$. A one-sided test against the directional alternative that the cross-sectional model is better would put every entry under 0.01, and the alternative is directional in substance, but we report the two-sided figure because it is the more demanding of the two. QLIKE, the harder loss to move, is significant at conventional levels throughout.

6.1 A caveat on the Clark-West column

Clark-West is a test about *models*. Its adjustment is derived on the null that the added coefficients are zero in population, and it adds back, in full, the estimation penalty the larger model pays for carrying them. That derivation assumes the larger model actually pays that penalty. Shrinkage removes part of it — that is what shrinkage is for — so if the two arms are shrunk by different amounts, the adjustment restores more than was ever charged and the surplus lands in the numerator of the statistic, inflating it. Nothing about that argument is specific to our data; it follows from how the correction in [11] is constructed. What is specific to our data is whether the precondition holds here.

It does. Our pipeline selects the ridge penalty on a validation slice, separately, for each arm, and the two arms disagree: they choose different penalties at 18% to 50% of refits depending on delay, and the cross-sectional arm is the more heavily penalised one more often than not — the configuration that would bias the statistic upward, which is the direction our column runs.

That is a reason to test the column, not to withdraw it, and the test is direct: impose a single ridge penalty on both arms, chosen on the restricted arm, so that the two shrinkages are equal by construction and the surplus has nowhere to come from. If the statistics barely move, the mechanism is not biting at this sample size, whatever it might do at another. We report alongside it the ratio of the mean Clark-West adjustment to the magnitude of the mean loss differential, which sits near one where the correction matches the penalty it removes and rises above one where it exceeds it. We offer that ratio as a description rather than a threshold: we have no external calibration for how large it must be before a p-value should be distrusted, and the common-penalty comparison is what the conclusion rests on.

Table 6. A diagnostic and a remedy for the Clark-West column, applied to this paper's own statistics. The ratio is the mean Clark-West adjustment over the magnitude of the mean loss differential; the common-penalty columns impose a single ridge penalty, selected on the domestic arm, across both.

δ	refits where the penalties differ	adjustment ratio	Clark-West	with a common penalty	Giacomini-White	with a common penalty
0	18%	1.10	0.23	0.28	-2.22	-2.17
3	50%	1.60	7.09	7.06	3.18	3.16
5	44%	1.27	8.02	8.01	4.48	4.45
13	46%	1.11	8.09	8.04	5.34	5.31
21	33%	1.03	8.01	8.02	5.66	5.65
55	33%	0.95	9.21	9.20	7.02	7.03

The diagnostic runs from 0.95 to 1.60, and only $\delta = 3$ exceeds 1.5 — the one Clark-West entry we would not lean on. The decisive column is the next one: imposing a common penalty moves no Clark-West statistic by more than 0.05, and changes no verdict at any delay. So the mechanism is present in principle and not detectable in practice here, and the column stands as a second opinion, reported with the ratio beside it rather than as a bare p-value.

The primary test is untouched, and for a reason rather than by luck. The argument above turns on Clark-West being a test about *models*, so that anything changing the estimation penalty changes what its adjustment ought to be. Giacomini-White is a test about *methods* under a fixed rolling window, and a method is the whole procedure including whatever tuning happens inside that window. The validation slice is carved out of the training window, so it is part of what is being tested rather than a contaminant of a correction — and Giacomini-White makes no finite-sample correction that could be thrown off. The prediction is that its column barely moves under the remedy, and the last two columns of Table 6 are that prediction tested: no statistic moves by more than 0.06 and no verdict changes at any delay. The substitution rates, the bootstrap intervals and the simultaneous bands involve no nested-model correction at all and are unaffected.

A second concern is that we test ten delays at once. Ten pointwise 95% intervals will breach somewhere by chance, so we bootstrapped the whole vector jointly, in moving blocks of ten days to respect the overlap in the target [18], and took the largest studentised deviation across the grid. The resulting critical value is 6.51 rather than 1.96 — a heavy penalty — and the effect still clears it from $\delta = 3$ to $\delta = 55$. One casualty is worth naming: the small negative reading at $\delta = 0$ does *not* survive the simultaneous band. We do not claim the cross-section is harmful when domestic data is current, and Section 5 gives the reason it is not — net of what it costs to estimate, its content there is zero.

The third concern is that the result is an artefact of fitting a wide model to a persistent series. We tested it by shifting the entire foreign block five years through the sample. The series keep their distributions, their autocorrelation, their crises; they lose only their alignment with the target. The substitution rate falls from +72% to -8%, and is negative at every delay we checked. Whatever the model is using, it is the alignment. The residual negative is not noise around zero and should not be read as one: it is the cost of breadth from Section 5, which is all that remains once the information is removed. A placebo of this kind lands below zero by construction, and a version of it that came back at exactly zero would be the surprising result.

Finally, Section 4 described the rate as slowing, which is a claim rather than an observation. Comparing $R(55)$ with earlier values: against $R(13)$ the difference is $+0.079$ with an interval of $[+0.037, +0.129]$, and against $R(21)$ it is $+0.042$ with $[+0.014, +0.076]$ — both still rising. Against $R(34)$ it is $+0.018$ with $[-0.001, +0.039]$, which does not reject equality. The rate is still climbing out to about seven weeks and flat thereafter.

One last property of the rate was assumed rather than tested. The rate is a ratio whose denominator is what the delay costs a domestic-only forecaster, and at short delays that quantity is near zero. A ratio with a denominator that can vanish is weakly identified, and a percentile bootstrap of it reports finite endpoints whether or not they mean anything. We therefore recomputed every interval by inverting a HAC t-test after Fieller, which returns two half-lines or the whole line when the data do not pin the ratio down. At every delay from one day to eleven weeks the confidence set is an ordinary interval, and Fieller's endpoints sit within 2.4 percentage points of the bootstrap's throughout — at $\delta = 55$, $[63\%, 79\%]$ against $[64\%, 78\%]$. The objection is right in principle and inert here, so the published intervals stand. Appendix C gives the construction, the one delay where the ratio does not exist at all, and the error that revealed the first version to be wrong.

7. The race against implied volatility

Everything to this point measures one candidate against a forecaster who has nothing else. Section 2 named the reason that is not enough. If the foreign cross-section is partly a slow reflection of what an options market reports immediately, then the substitution we measured is real and the instrument is wrong, and a practitioner following our advice would buy the expensive thing.

So we put the two candidates in the same walk-forward, on the same labels, with the same estimator. Four models: the domestic block alone, the domestic block plus foreign closes, the domestic block plus implied volatility, and everything together. VDAX-NEW begins in May 2001 against the index panel's February 2000, which costs 6% of the test days, so all four are re-estimated on the same reduced window of 4,525 days and the comparison stays internal.

Table 7. Out-of-sample R^2 for four information sets, 4,525 days, September 2008 to September 2026. The last two columns are what the foreign block adds *given* implied volatility, and the Giacomini-White statistic testing it. VIX enters at $t - 1$, the strictly admissible specification of Section 3; the same-day timing sensitivity check follows in the text.

δ	domestic	+ foreign	+ implied vol	+ both	foreign IV	GW z
0	0.4958	0.4865	0.5379	0.5272	-0.0108	-2.48
3	0.3822	0.4270	0.5162	0.5053	-0.0109	-2.20
5	0.3350	0.4221	0.5151	0.5053	-0.0098	-1.92
13	0.1990	0.3928	0.5083	0.4989	-0.0093	-1.69
21	0.1082	0.3751	0.5068	0.4968	-0.0100	-1.81
55	-0.0369	0.3534	0.5088	0.4990	-0.0098	-1.73

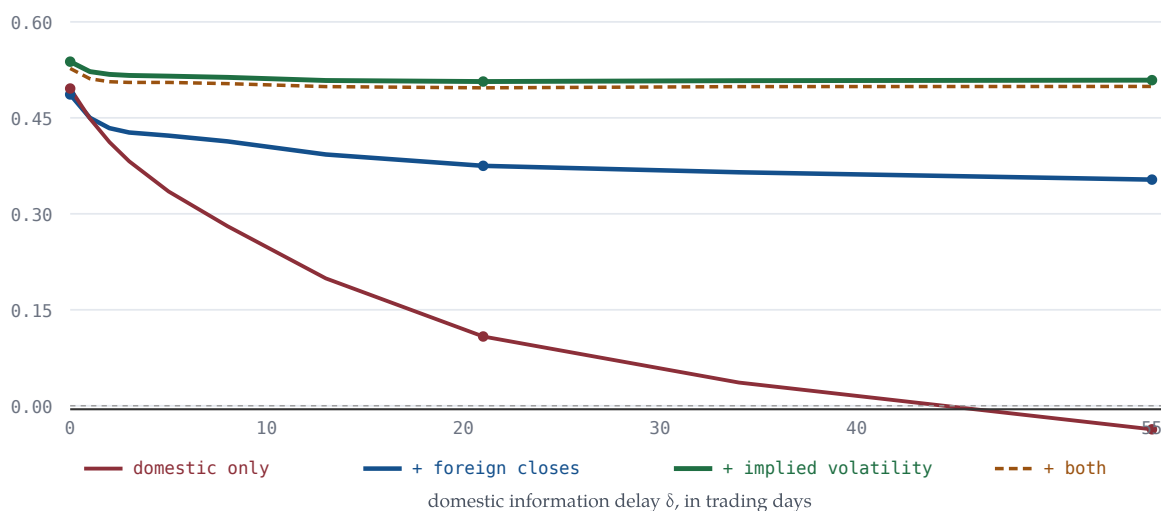


Figure 2. The green curve is almost flat. A forecaster holding implied volatility barely notices how old their domestic data is, and from about four days on they are ahead of where the domestic-only forecaster started. The dashed curve, which adds foreign closes on top, sits just below it at every delay.

The green curve is the result. Implied volatility does not merely slow the decay, it very nearly stops it: R^2 falls from 0.538 to 0.509 across eleven weeks, against 0.496 to -0.037 for the domestic model alone. Expressed as a substitution rate that is 103% at three weeks, and a rate above 100% is the signal from Section 3 that the measure has changed meaning. A forecaster holding three-week-old domestic data and today's VIX scores 0.507, above the 0.496 of a forecaster with fully current domestic data and no options market. Implied volatility is not restoring what staleness took; it is supplying something the realised-variance history never contained.

That should not be mysterious, though it should not be overstated either. The VIX is a RISK-NEUTRAL expectation of S&P variance over roughly the coming MONTH; our target is REALISED variance over the coming five days. The two differ in horizon, by a factor of about four, and in measure, by a variance risk premium that is itself time-varying. So the VIX is not the quantity we are predicting. It is, however, a purpose-built forward-looking forecast of something close to it, and comparing that with a cross-section of correlated state variables is comparing a forecast with a set of clues. Read that way the surprise is not that implied volatility wins but that seven foreign equity markets get within two-thirds of it despite the horizon and measure mismatch working against the options market.

The sixth column is what the objection actually turns on, and it is negative at every delay. Given implied volatility, adding the foreign block makes the forecast slightly worse, and the Giacomini-White statistic — the domestic-plus-IV model is nested inside the full one, so the same test applies as in Section 6 — reaches conventional significance at the short delays. Taken literally this says foreign closes damage a forecast that already has VIX, which would be strange behaviour for genuine information.

Section 5 supplies the alternative reading, and the same instrument settles it. Replacing the seven foreign series with AR(1) surrogates matched on persistence and variance leaves regressors that carry nothing by construction, and what remains is the bill for estimating them. It comes to roughly 0.011 to 0.015 of R^2 , which is the whole of the negative. The content of the foreign block given implied volatility, gross of that cost, is $+0.001$ at $\delta = 0$, $+0.007$ at $\delta = 5$, $+0.008$ at $\delta = 21$ and $+0.007$ at $\delta = 55$,

and bootstrapping each of those the way Section 5 describes gives intervals that all cover zero: $[-0.007, +0.010]$, $[-0.002, +0.017]$, $[-0.002, +0.019]$ and $[-0.003, +0.017]$. The correct word is redundant. Foreign closes do not damage a forecast that already sees the options market; they add nothing to it, and paying to estimate nothing is what makes the measured figure negative.

The table above is the admissible specification: VIX lagged a full day, which discards a day of information the live index would have carried. Substituting the same-day close, which contains fifteen minutes we are not entitled to, moves the options market the other way: R^2 at $\delta = 21$ rises from 0.507 to 0.568, a substitution rate of 119%, while the incremental contribution of the foreign block stays negative at all ten delays and its gross content stays indistinguishable from zero. Neither arm is the specification we would choose, and the one we would choose is not available to us. What licenses a conclusion here is not that the 21:00 observation falls between these two in time — that would say nothing about how it forecasts — but that the verdict is the same under a specification that gives the options market a full day less information and under one that gives it fifteen minutes more. A finding that survives that much variation in the input is not resting on the timing choice.

7.1 Redundancy given VIX and VDAX, with one regressor instead of seven

The test above adds seven regressors to the implied-volatility model, and a referee points out that it therefore cannot separate two different reasons for a null: the foreign block carries nothing beyond VIX and VDAX, or it carries something and the cost of estimating seven coefficients consumes it. Section 5 says that cost is real and Section 5.1 says a compression avoids part of it, so the confound is not hypothetical. The compression supplies the fix — add *one* regressor instead of seven.

Table 8. What a single compressed summary of the foreign block adds on top of stale domestic data and the implied-volatility block, which is VIX and VDAX-NEW. The mean requires no estimation at all; PC1 is the real-time first component of Section 5.1. The seven-regressor column is Table 6's, for comparison.

δ	+ IV only	+ mean	+ PC1	+ all seven	mean IV	GW z	PC1 IV	GW z
0	0.5379	0.5392	0.5391	0.5272	+0.0012	0.50	+0.0012	0.48
3	0.5162	0.5184	0.5187	0.5053	+0.0022	0.98	+0.0025	1.07
5	0.5151	0.5179	0.5183	0.5053	+0.0028	1.02	+0.0032	1.12
13	0.5083	0.5120	0.5124	0.4989	+0.0037	1.16	+0.0042	1.26
21	0.5068	0.5106	0.5110	0.4968	+0.0038	1.19	+0.0042	1.28
55	0.5088	0.5129	0.5130	0.4990	+0.0041	1.31	+0.0043	1.30

The sign flips and the conclusion does not. With seven regressors the increment is negative at every delay; with one it is positive at every delay. That is the estimation-cost mechanism of Section 5 showing up exactly where it was predicted to, and it means the negative in Table 6 was a bill rather than evidence of harm — a reading the paper offered there and can now demonstrate rather than argue. But no Giacomini-White statistic reaches 1.96: the largest is 1.31, at eleven weeks. A single cheap summary of seven foreign markets, carrying essentially no estimation cost, still adds nothing detectable to a forecaster holding implied volatility.

That is a stronger statement than the one this section made before, and it should be read as strictly stronger rather than as a reversal. The earlier test could be dismissed as penalising the foreign block for its parameter count. This one cannot. What it cannot do either is prove absence: increments of +0.001 to +0.004 with statistics between 0.48 and 1.31 are consistent with a small positive

contribution this sample is too small to resolve, and the honest summary is that the foreign block adds nothing we can detect given implied volatility, not that it adds nothing.

7.2 Is redundancy an artefact of the linear estimator?

Every model above is a ridge regression, so the conclusion so far is that the foreign block adds nothing *linearly*. The objection writes itself: a non-linear map might find cross-market structure that bypasses the single global factor, in which case the redundancy is a property of the estimator and not of the information. The same question was already put to the domestic-plus-foreign feature set: squaring every regressor makes the forecast worse at five of six delays, and letting the global factor enter quadratically and interact with the domestic state moves nothing that reaches significance in either direction. That result never sees implied volatility, though, so it does not answer this one. The two are independent: a non-linear foreign map could be redundant with the linear one and still add something on top of VIX.

Three non-linear arms are therefore added on top of the domestic-plus-implied-volatility model. FACTOR carries the global mean g of the foreign block with g^2 , an interaction with the domestic state, and an interaction with VIX — the last being the term that would show if foreign closes mattered only when the options market was already nervous. The other two replace g with 20 and 60 random Fourier features of the foreign block. A random Fourier layer, $z(x) = \sqrt{(2/m)} \cos(Wx + b)$ with W and b drawn once from fixed distributions, is a one-hidden-layer network with a random first layer and a ridge-fitted output layer, and approximates a Gaussian RBF kernel as m grows; the map it represents is neither polynomial nor limited to any interaction order. Its scaling and bandwidth come from the burn-in period alone, so no test day contributes to them.

Two details decide whether such a test is fair, and both cut against the answer we happen to get. First, the widest arm carries 72 regressors, and Appendix A reports validation already choosing the top of the penalty grid most of the time with ten. A grid that stops at 30 would force the flexible arms to be under-regularised, and they would then lose for a reason unrelated to information. Every comparison is therefore run twice, once on the paper's grid and once on a grid extended thirtyfold; the verdict is read from the wider one. Second, on the wider grid the implied-volatility baseline itself is slightly *weaker* — R^2 at $\delta = 21$ falls from 0.507 to 0.494 — so the target the foreign arms must beat has been lowered, not raised.

Table 9. What each non-linear map of the foreign block adds to a forecaster holding stale domestic data and implied volatility. Wider penalty grid; Giacomini-White statistics, the implied-volatility model being nested in each. A statistic above +1.96 would overturn the redundancy result.

δ	R^2 IV	+ foreign	GW z	+ FACTOR	GW z	+ RFF(20)	GW z	+ RFF(60)	GW z
0	0.5339	−0.0097	−2.47	−0.0111	−2.70	−0.0414	−2.95	−0.0538	−3.78
3	0.5066	−0.0051	−1.02	−0.0097	−1.84	−0.0457	−2.27	−0.0620	−3.19
5	0.5051	−0.0057	−1.04	−0.0076	−1.27	−0.0548	−2.48	−0.0723	−3.47
13	0.4955	−0.0054	−0.82	−0.0098	−1.02	−0.0652	−2.96	−0.0766	−3.49
21	0.4943	−0.0047	−0.79	−0.0161	−1.46	−0.0704	−2.83	−0.0830	−3.59
55	0.4903	−0.0048	−0.79	−0.0047	−0.70	−0.0614	−2.62	−0.0686	−2.91

No arm adds anything at any delay, and the two random-feature arms are significantly *worse* at all six. The direction of the capacity comparison is the informative part: RFF(60) is below RFF(20) at every

delay, by 0.007 to 0.018 of R^2 . If the objection were right — if there were cross-market structure a linear map could not reach — more flexibility would find more of it. More flexibility finds less, which is what over-parameterisation looks like when the signal is not there. The surrogate measurement puts a number on the mechanism: bending the foreign block costs 0.005 to 0.008 of R^2 for the FACTOR arm and 0.017 to 0.056 for the random-feature arms, rising with both delay and width.

One thing genuinely changes under the wider grid, and it favours the objection rather than us. The *linear* increment, significantly negative at short delays in Table 7, stops being significant from $\delta = 3$ onward, with statistics between -0.79 and -1.04 . That is the right result and the one Section 5's estimation-cost argument predicted: the foreign block given implied volatility is not harmful, it is empty, and the apparent harm in Table 7 was the bill for estimating regressors that carry nothing. The redundancy conclusion is unchanged; its sign is better explained.

What this does not license is the claim that no non-linear model could find anything. The random first layer is not learned, and a learned one has strictly more free parameters — so it faces the same estimation-cost bill this sample has repeatedly failed to pay, which is an argument by extension rather than a test. Section 9 records it as such.

7.3 Is five days the result, or the setting?

Every number above forecasts five trading days ahead, and a reader is entitled to ask whether that is a finding or a convention. So the whole experiment is re-run at one day, two weeks and one month. One detail has to be settled before the table can be read: the Yang-Zhang proxy spans five days, which is how the latent variance is measured and has nothing to do with how far ahead we forecast, but it means that at $h = 1$ the outcome window shares four of its five days with the domestic feature the model already holds. Nothing leaks — the outcome is still strictly in the future and every feature is still dated $t - \delta$ or earlier for the domestic block — but a forecaster at one day is being asked to predict an object they can largely already see, and their R^2 at $\delta = 0$ duly comes out at 0.921 against 0.499 at five days. The serial correlation of the target confirms it: 0.953 at one day against 0.692 at five, 0.576 at ten and 0.417 at twenty-one.

Table 10. The substitution rate at eleven weeks under four forecast horizons. Intervals are the full-ratio bootstrap of Section 4, in moving blocks of twice the horizon. The $h = 5$ row is the paper's headline and reproduces it exactly.

horizon	own R^2 at $\delta = 0$	$R(55)$ under R^2	95% CI	$R(55)$ under QLIKE
1 day	0.921	70%	[67%, 72%]	83%
5 days	0.499	72%	[64%, 79%]	81%
10 days	0.338	66%	[52%, 78%]	73%
21 days	0.161	51%	[23%, 71%]	65%

From one day to two weeks the rate is flat — 70%, 72%, 66% — across a tenfold change in horizon and despite the one-day column measuring something partly mechanical. At one month it falls to 51%, the only point estimate outside the interval the headline quotes, though the intervals still overlap heavily and the one-month interval is wide enough on its own to be uninformative. QLIKE tells the same story more smoothly: 83%, 81%, 73%, 65%. So the headline is a property of the information over horizons out to a fortnight and begins to decay beyond that, which is what should happen — a foreign close carries information about the next week's volatility state more readily than about the next month's, and by twenty-one days the domestic model it is being compared against has

itself fallen to an R^2 of 0.161, so there is less loss left to recover and a noisier denominator to recover it against. We report the five-day figure as the headline because it is the horizon the whole paper is built on, not because it is the most favourable: at one day under QLIKE the rate is higher.

7.4 What the R^2 column means to someone sizing a position

R^2 and QLIKE establish that something is there; neither says whether it matters. A variance forecast is used to size something, so the quantity that matters is how far wrong it is as a *multiple* of the truth. Taking the same forecasts the paper has already produced, applying the smearing correction of Section 6 so the variance forecast is mean-unbiased, and reading off the ratio of forecast to realised variance: at $\delta = 0$ both models sit outside a factor of two on much the same share of days, 31.1% for the domestic model and 31.4% for the cross-sectional one, and at eleven weeks the stale domestic model does so on 59.2% against the cross-sectional model's 37.1%. That 22-point reduction is the R^2 column in units a risk manager can act on. The floor is the first thing to notice and it is high: three days in ten are outside a factor of two even with fully current data, because five-day realised variance is a noisy object.

The second thing is more important and is not what we expected. Splitting that tail shows the stale model does not fail by *under*-forecasting. It fails by over-forecasting, its median ratio drifting from 1.26 at $\delta = 0$ to 1.90 at eleven weeks while the cross-sectional model's stays put at 1.27, and almost the whole 22-point improvement is the cross-section pulling that overstatement back. The understating tail — forecasts below half the realised variance, which is the side that leaves a position too large — runs between 8.6% and 11.4% of days and is essentially unmoved by breadth at every delay, changing by less than a percentage point and in the wrong direction at three of the four we checked. A practitioner adopting the cross-section to avoid being caught short by a volatility spike would be adopting it for something it does not do. What it does is stop a stale forecaster from systematically over-estimating risk, which is a real cost and a different one. None of this is a trading result: there is no strategy, no transaction cost, and no claim that any of it survives contact with an execution desk.

7.5 Would a richer options block change the answer?

Section 7 runs on two implied-volatility series. The obvious question is whether redundancy is an artefact of that thinness, and the obvious test is to add a third. We obtained the spot VSTOXX index for that purpose, and report here both what it shows and why it does not become part of the headline.

It does not become part of the headline because requiring it costs 73% of the sample. The export runs from December 2012 to March 2025, and the walk-forward needs 252 + 1,250 + 250 days of burn-in plus the longest delay before it can produce a single test day, so insisting that all three series be present moves the test window from 4,525 days beginning September 2008 to 1,240 days beginning April 2020. That discards the 2008 crisis outright and begins *after* the March 2020 spike: peak VIX is 82.7 over the full sample and 41.4 inside what remains. A volatility study that has dropped every major volatility episode in its own sample is not a stronger one.

What can be done is to hold the days fixed and vary only the block, which separates the third series from the shorter, calmer window — and the window is by far the larger effect. On the same 1,240 days, the two-series options arm scores 0.4386 at $\delta = 0$ and 0.3952 at eleven weeks, against 0.5379 and 0.5088 on the full sample; almost all of that difference is the window, not the block.

Table 11. The third implied-volatility series, on the 1,240 days where all three exist. IV2 is VIX and VDAX; IV3 adds spot VSTOXX. The last two columns are what the seven foreign closes add on top of IV3, with IV3 nested inside the full model.

δ	+ IV2	+ IV3	IV3 – IV2	GW z	foreign IV3	GW z
0	0.4386	0.4348	–0.0039	–1.27	–0.0063	–0.60
3	0.4024	0.3974	–0.0050	–1.24	–0.0216	–2.08
5	0.4008	0.3947	–0.0061	–1.60	–0.0241	–2.37
13	0.3936	0.3864	–0.0072	–1.96	–0.0243	–2.47
21	0.3901	0.3824	–0.0077	–1.94	–0.0237	–2.40
55	0.3952	0.3886	–0.0065	–1.76	–0.0193	–1.78

The third series adds nothing at any delay and is mildly negative at all six. That is not surprising once stated: VSTOXX and VDAX are both European equity volatility indices, so the third column is close to collinear with the second and costs more to estimate than it returns — the same arithmetic Section 5 applies to the foreign block. It is, however, the opposite of what an earlier draft of Section 9 asserted would happen, and that assertion is withdrawn there.

The question the test was built for is answered in the last two columns: given all three options series, the foreign block still adds nothing, at every delay. The redundancy result is not an artefact of a thin options block. Two cautions belong with that. The incremental figures are negative rather than zero, and Section 5's estimation-cost argument applies here with more force than it does in Table 7, since the full model now carries ten options and foreign regressors on a fifth of the days — the negative is a bill, not a harm. And 1,240 quiet days support a weaker “adds nothing” than 4,525 days spanning two crises do. What this cannot speak to at all is 2008 to 2012, because the data needed to add the third series does not reach back that far.

8. Where each result holds

Repeating the breadth exercise with each of five indices as the target, gives the following at $\delta = 55$. An earlier draft ran this check on the binary target, for comparability with the companion note, which was inconsistent: if a median split discards the magnitude information a cross-sectional model uses, as that note argues, then the external-validity check was running on a weaker measurement than the result it supports. It is repeated here on the continuous target.

Table 12. Substitution rate at eleven weeks, by target market, on the CONTINUOUS target and the same losses as Table 1. Same-day peers are markets that close earlier on the same calendar day; the remainder contribute the previous session. Intervals bootstrap the whole ratio.

target	close, UTC	same-day peers	R ² own, $\delta=0$	R ² own, $\delta=55$	R ² cross, $\delta=55$	R(55)
S&P 500	21:00	6	0.4987	–0.0351	0.3470	72% [64, 79]
FTSE 100	16:30	4	0.4573	–0.0112	0.4003	88% [80, 94]
DAX	16:30	4	0.5063	0.0117	0.3774	74% [67, 80]
Hang Seng	08:00	2	0.4237	–0.0376	0.2813	69% [58, 80]
Nikkei 225	06:00	0	0.3374	–0.0313	0.1886	60% [45, 73]

The effect appears everywhere, and the Nikkei is the case that carries information. It closes before every other market in the sample, so it has no same-day foreign observation at all — every peer reaches it a session late — and it recovers the least, 60% against 69% to 88% elsewhere. That is what the mechanism predicts and what a spurious correlation would not produce. Past that contrast we can say little: the S&P has six same-day peers and recovers less than the FTSE with four, the intervals overlap heavily, and five markets cannot resolve an ordering.

One further objection deserves an answer. Rapach, Strauss and Zhou find that lagged US returns predict foreign returns while the reverse is weak [16]. That concerns returns rather than volatility, and lagged returns rather than same-day closes observed before the target's own close. We rely on both distinctions, and a reader who rejects them should discount Sections 4 through 6 accordingly.

The scope condition matters more than the ordering. Section 7 shows that where a liquid options market exists on the asset, breadth is redundant, and the S&P 500 is the most heavily optioned index in the world. So the breadth result may be most relevant to assets without such a market, which is the case Section 1 opens with: a private portfolio marked quarterly, a property book on an appraisal cycle, a fund reporting NAV with a lag. Nobody writes a liquid option on those. What a holder of them can see is the public cross-section, and Sections 4 through 6 say what that is worth as a function of how stale the mark has become.

We would rather state that plainly than dress it up, because it is also the weakest point of the design. We ran the experiment on a liquid asset in order to observe the truth, and the property that makes the truth observable — a deep, continuously traded market — is the same property that brings an options market with it. The result we can measure cleanly and the case we care about are therefore separated by the one feature that decides which substitute wins. Section 9 says what would close that gap.

There remains the $\delta = 0$ row of Table 1, and it is the one place in this paper where the choice of loss decides the sign. On log-scale squared error the cross-section is worse by 0.0103, interval $[-0.0192, -0.0016]$; on QLIKE it is worse by 0.0142, $[-0.0292, -0.0023]$; on squared error on the variance scale it is BETTER by 0.0071, $[-0.0470, +0.0280]$. Two of the three exclude zero and point one way, the third does not exclude zero and points the other. An earlier draft read the negative as settled and used it to press a disagreement with Korkusuz [9] and Jayawardena et al. [6], who find foreign volatility improves a current forecast. We withdraw that. A sign that moves with a monotone transformation of the target is not a finding about the world, and the two losses that agree are the log-scale one, which is not proxy-robust, and QLIKE, which shares the log-ratio structure. The honest statement is that at zero delay our design cannot tell which way the effect runs, and nothing here bears on their result. Table 2's decomposition still holds conditional on the log scale: whatever is there is small, bounded near one point of R^2 , and mostly the bill for estimating seven coefficients.

9. What we cannot say

The limitation that matters most is the one Section 8 names. The illiquid-asset case that motivates the whole exercise is argued rather than demonstrated, and Section 7 raises the cost of that gap rather than leaving it where it was: we now know the answer depends on whether an options market exists on the asset, and we have tested only the side where one does. Section 11 narrows the gap by one step, showing that the clean-lag model of staleness behaves like periodic marking at matched mean staleness, so the apparatus is at least the right shape for the case we cannot reach. It does not close it.

Only a genuine appraisal-based or quarterly-marked target, on an asset with no options written on it, would do that.

The implied-volatility block is thinner than we would like. VIX and VDAX cover the sample. A spot VSTOXX index was subsequently obtained, and Section 7.4 reports what it shows; the short version is that it does not cover the sample either, so the headline block stays at two series. An earlier draft asserted here that a fuller block “might widen the gap in Section 7 further” and “would not narrow it.” That assertion was wrong in direction and is withdrawn: on the window where three series are available, the third one makes the options arm slightly worse, because VSTOXX and VDAX are both European volatility indices and the third column costs more to estimate than it returns. The conclusion it was defending — that the foreign block is redundant given implied volatility — is unaffected, which is why the error was worth correcting rather than quietly dropping.

The non-linearity result in Section 7.2 is bounded in a specific way. A random Fourier layer is a one-hidden-layer network whose first layer is drawn rather than learned, and no network with a learned first layer was fitted. The argument that one would not change the answer is an extension of the estimation-cost finding — more free parameters, the same bill, a sample that has repeatedly failed to pay it, and a capacity comparison already pointing the wrong way — and an extension is not a test. Section 7.4 names a limitation of a different kind, and it is the one a practitioner should read first: what breadth repairs is a stale forecaster's tendency to over-estimate variance, not their exposure to under-estimating it. The understating tail is unmoved at every delay we checked. Anyone reaching for the cross-section as protection against being caught short by a volatility spike is reaching for the wrong thing. Beyond that: nothing here bounds what a better estimator could do. An earlier draft said at this point that a cheaper representation of the foreign block *would* recover part of what estimation costs, which was an inference from a counterfactual generator rather than a result. Section 5.1 now fits that representation instead, and it does recover part of the cost — but the reader should note that the paper asserted the conclusion before it had earned it. The horizon evidence in Section 7.3 runs out at one month, where the rate has already fallen to 51% with an interval too wide to lean on; we do not know what happens at a quarter. Equity indices only. Close times are treated as standard, ignoring daylight saving, which moves them by an hour and never reorders the blocks. The price data are a retail export with no point-in-time guarantee. A referee asks for an official source and an exactly timestamped VIX, and the two requests have different answers. The official CBOE history is public, and substituting it for the retail VIX export would retire the point-in-time concern for that one series; we have not done it here and flag it as the obvious next revision. It would not, however, supply the observation the section above actually wants, because that file carries one open, high, low and close per day and no intraday snapshot at all. The 21:00 value of VIX existed and is the right regressor; it is not in any daily product, official or retail, and the lagged and same-day arms bracket it rather than recover it. Differences at $\delta \leq 2$ fall inside the simultaneous bands. The variance-scale loss is dominated by a handful of crisis days, which is why its intervals are wide enough to be uninformative at short delays.

10. Reproducing this

Every number above is printed by one of twenty scripts — `lab03_crossection.py`, `lab04_walkforward.py`, `lab05_robustness.py`, `lab06_inference.py`, `lab07_estimation_cost.py`, `lab08_implied_vol.py`, `lab09_nonlinearity.py`, `lab10_loss_scale.py`, `lab11_markets_continuous.py`, `lab12_appendix.py`, `lab13_origin_median.py`,

lab14_appraisal_smoothing.py and lab15_nonlinear_given_iv.py and lab16_clark_west_shrinkage.py, lab17_horizons.py and lab18_economic_reading.py and lab19_vstox_third_series.py, lab21_stronger_inference.py and lab22_factor_benchmark.py and lab23_compressed_everywhere.py — run on the data of Section 3 under a fixed seed. They are collected at github.com/SunnyAlexV/delayed-information-cross-section, together with the data, the reference output and both papers. A single command, `python run_all.py --check`, reruns every script and compares each line against that stored output. Four of those scripts were additionally reproduced on a second machine running a different operating system and matched digit for digit; the rest have been run only on the platform that generated the reference output, and we state that rather than the stronger claim. The seam audit prints on every run, including the seam it could not verify. Nothing is reported that the scripts do not produce.

That command establishes only that the scripts still print the same numbers. It says nothing about whether this paper still quotes them, and twice during drafting it did not: a batch of edits aborted partway, discarding replacements that had already been applied, and a superseded figure survived into a draft. A second command, `python verify_paper.py`, closes that gap by reading the source of both papers and checking every cost, gross, interval and test statistic in them against the stored output, together with a list of superseded values that must not reappear and withdrawn claims that must not return. It makes two hundred and sixty-one such checks and exits non-zero on any disagreement.

11. Does a clean lag resemble a stale mark?

Section 12 will argue that what carries from this paper to an illiquid asset is the apparatus rather than the numbers. One part of that claim can be checked rather than hoped for, and this section checks it. Every result here models staleness as a clean lag: the forecaster sees the state exactly as it stood δ days ago and nothing since. A marked asset is not stale in that way. It is revised on a schedule and held flat in between, and the revision is a partial adjustment toward current evidence rather than a full mark, which is how the property literature has modelled appraisal indices for thirty years [8]. If the substitution rate under that kind of staleness looked nothing like the rate under a clean lag, $R(\delta)$ would be a curve about clean lags alone.

It does not. Rebuilding the domestic block as a mark revised every q days and held between revisions, and matching on mean staleness, gives 64% against the clean lag's 60% at ten days, 71% against 70% at thirty-one, and 70% against 72% at ninety-four. Adding partial adjustment on top moves it a few points the other way, 63% where matching predicts 70%, so the equivalence is approximate and we claim no more than that. One thing we got wrong on the way is worth recording: our first attempt modelled appraisal as pure exponential smoothing, and that is not staleness at all. A weighted average of the recent past still contains today's observation, so it costs almost nothing — R^2 falls from 0.499 to 0.456 across a mean lag of eleven weeks, against 0.499 to -0.035 under a clean lag — and leaves almost no damage for breadth to repair. Holding a number flat between revisions is what makes a mark stale, not blurring it.

12. What to do with this

Someone holding a stale mark on an asset with a liquid options market should buy the options data. Section 7 provides strong evidence for that in this setting, and the foreign cross-section adds nothing

measurable once implied volatility is in hand.

Someone holding a stale mark on an asset with no options market, which is the case Section 1 opens with, has breadth as the only candidate on the table. Sections 4 to 6 show how that worth can be measured and give a liquid-market benchmark for it, rather than an estimate for the illiquid asset itself: roughly half the delay-induced loss recovered within a week, around 70% by eleven weeks, against a cost of one to five points of R^2 for carrying the extra series.

One generalisation is worth refusing explicitly. It would be convenient to present Figure 1 as a benchmark for how quickly information decays in delayed systems, and it is not one. Those curves are a measurement of one index against seven specific peers over one sample. The correlation between a private portfolio and the public cross-section is a different quantity, and nothing here bounds it. A reader who wants a number for their own asset will not find it in this paper.

What does carry across is the apparatus rather than the numbers. $R(\delta)$ is defined in units that survive a change of loss function, market or candidate. The delay sweep is a design anyone with a delayed series and a candidate substitute can run, and the horse race in Section 7 is the question to ask first, because its answer decides whether breadth is worth buying at all. The cost decomposition in Section 5 separates what a source knows from what a forecaster keeps, which is the distinction that turned our own most confusing result into an ordinary one. Pointed at an appraisal-based or quarterly-marked series, that machinery would produce the numbers this paper cannot — and the check above says the delay model it carries is the right shape for that series, even though the numbers are not.

Appendix A. Settings, and what they are worth

Three choices below decide numbers in the paper and are stated nowhere else, so they are stated here, together with a test of the two a reader is most entitled to worry about.

The estimator uses a rolling 1,250-day training window with a 250-day validation tail, refitted every 21 days, with decision variables formed against a trailing 252-day median. Standard errors on loss differentials are Newey-West with a Bartlett kernel at nine lags, which is the usual bandwidth rule at 4,862 observations. Bootstraps use 2,000 replicates in moving blocks of ten days, twice the forecast horizon. The ridge penalty is chosen from $\{0.3, 1, 3, 10, 30\}$ on the validation tail.

That last choice looks worse than it is. Validation lands on an endpoint most of the time: at $\delta = 21$ the cross-sectional model picks the largest penalty in 84% of refits and the smallest in 14%, which would ordinarily mean the grid is choosing the model. The test is not where selection lands but whether widening the grid moves the answer, and widening it thirty-fold in each direction, to $\{0.03, \dots, 1000\}$, moves ΔR^2 by 0.008 at five days and 0.003 at eleven weeks — in the cross-section's favour. The penalty surface is flat enough that the endpoint selection is cosmetic. We report the grid because a reader should not have to take that on trust.

Block length is the other one, and it does some work. Across blocks of 5, 10, 21 and 42 days the point estimate is unchanged by construction and every interval excludes zero, so the conclusion does not depend on the convention. The width does: at $\delta = 55$ the interval runs $[+0.332, +0.431]$ at a five-day block and $[+0.285, +0.474]$ at forty-two, about 40% wider. Longer blocks carry more dependence and should give wider intervals, so the two-month block is the conservative reading and the ten-day intervals quoted throughout are not the most cautious available. Nothing in the paper turns on the

difference, but a reader comparing our intervals with another study's should know which end of that range ours came from.

Appendix B. How the null generator was got wrong, twice

Section 5.2 measures the cost of breadth a second time with an unfitted generator: the real foreign block moved circularly, so that every tail and cluster is the real one and only the alignment is destroyed. Two traps sit inside that sentence. An earlier version of the measurement fell into both, and reported a disagreement between the two generators that was neither generator's doing. The first is that the gaps move with the values. The foreign block is complete on 96.2% of rows; the remainder are holidays that differ across markets. Rolling the raw array moves that missingness pattern too, so on about 187 test days per draw — 3.8% — the surrogate arm has no usable row where the real arm does, and it loses the matching training rows as well. Both push the measured cost down, and neither is a cost of breadth. The NAIVE column is that mistake left in: it sits 0.0207 of R^2 below the corrected estimate at $\delta = 0$, where the real cost is small, and 0.0011 below it at $\delta = 55$, where the real cost is large. A penalty that is nearly constant in absolute size while the quantity being measured grows more than fivefold is a fixed charge for lost days, not a cost of carrying regressors. The fix is to shift the values while holding the real mask fixed, which is what the SHIFT column does.

The second trap is that an offset can be a near-identity. Offsets are taken modulo the sample length, so an offset a few rows short of the full length is a shift of a few rows *backwards*: the real block, barely moved, still carrying the signal. One such draw in eight put the mean cost at +0.0140 with a standard deviation of 0.0709, against the -0.0122 the guarded grid gives; the offending offset was a shift of eight rows, and scored +0.1893 on its own. Every offset here is held at least two hundred rows from identity in either direction, and the guard is printed in the lab output rather than assumed.

Neither trap is deep, and that is the point worth taking from this appendix. A null generator is a piece of code, and a placebo that quietly stops being a placebo returns a plausible number with no outward sign of trouble. The reason we can say the bill is not an artefact of the Gaussian assumption is that two generators with almost nothing in common agree on it; the reason we can say that at all is that the two ways of getting the unfitted generator wrong were found and fixed first.

Appendix C. A confidence set for a ratio whose denominator can vanish

The substitution rate is a ratio, and Section 6 reports that inverting a HAC t-test after Fieller leaves every published interval standing. This is how that was computed, and why it was not the first answer we got. Fieller's theorem inverts a t-test on the linear combination num – R·den, which is zero at the true rate, and solves the resulting quadratic for the set of rates not rejected. When the denominator is well separated from zero the solution is an ordinary interval close to the delta-method one. When it is not, the quadratic's leading coefficient changes sign and the confidence set is correctly a union of two half-lines, or the whole line — an honest statement that the data do not pin the ratio down, and one a percentile bootstrap cannot make. We have met the symptom in this project without naming it, in an interval of $[-154\%, +50\%]$ at one point, and treated it as a nuisance rather than as the identification problem it is. Two things about the calculation are worth recording. The variances must be HAC, with the same Bartlett kernel used in Section 6, because the five-day targets overlap: the first version used iid variances and returned intervals *narrower* than the block bootstrap at every delay, which is impossible for overlapping targets and was the only sign that anything was

wrong. And $\delta = 0$ is absent from the comparison throughout, because there the denominator is identically zero by construction — the one delay at which the ratio does not exist at all, rather than being merely hard to pin down.

References

1. Buncic, D. and Gisler, K. I. M. (2016). Global equity market volatility spillovers: a broader role for the United States. *International Journal of Forecasting* 32(4), 1317–1339.
2. Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7(2), 174–196.
3. Engle, R. F., Ito, T. and Lin, W.-L. (1990). Meteor showers or heat waves? Heteroskedastic intra-daily volatility in the foreign exchange market. *Econometrica* 58(3), 525–542.
4. Diebold, F. X. and Yilmaz, K. (2014). On the network topology of variance decompositions: measuring the connectedness of financial firms. *Journal of Econometrics* 182(1), 119–134.
5. Goetzmann, W. N., Ivković, Z. and Rouwenhorst, K. G. (2001). Day trading international mutual funds: evidence and policy solutions. *Journal of Financial and Quantitative Analysis* 36(3), 287–309.
6. Jayawardena, N. I., Todorova, N., Li, B. and Su, J.-J. (2020). Volatility forecasting using related markets' information for the Tokyo stock exchange. *Economic Modelling* 90, 143–158.
7. Getmansky, M., Lo, A. W. and Makarov, I. (2004). An econometric model of serial correlation and illiquidity in hedge fund returns. *Journal of Financial Economics* 74(3), 529–609.
8. Geltner, D., MacGregor, B. D. and Schwann, G. M. (2003). Appraisal smoothing and price discovery in real estate markets. *Urban Studies* 40(5–6), 1047–1064.
9. Korkusuz, B. (2025). Beyond the S&P 500: examining the role of external volatilities in market forecasting. *Review of Economic Design*. DOI 10.1007/s10058-024-00373-x.
10. Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica* 74(6), 1545–1578.
11. Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics* 138(1), 291–311.
12. Diebold, F. X. (2015). Comparing predictive accuracy, twenty years later: a personal perspective on the use and abuse of Diebold-Mariano tests. *Journal of Business & Economic Statistics* 33(1), 1–9.
13. Vellankaran, S. A. (2026). Why a simple threshold rule is hard to beat: volatility regime prediction under information delay. Working note.
14. Yang, D. and Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business* 73(3), 477–491.
15. Zhao, X., Qiao, H. and Lin, Z. (2026). Forecasting global volatility across asynchronous markets: incremental accuracy from constrained cross-market attention. arXiv:2608.25369v1. Superseded by v2 (Zhao and Qiao, 5 September 2026) under a different title.
16. Rapach, D. E., Strauss, J. K. and Zhou, G. (2013). International stock return predictability: what is the role of the United States? *Journal of Finance* 68(4), 1633–1662.
17. Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics* 160(1), 246–256.
18. Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* 17(3), 1217–1241.